Provability closures for the quantifiers

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs ax:q1, ax:q2; note the distinct label thm:provability-quantifiers (source line 1420) is a different statement whose prvEx item is an Exercise and is rejected.

Editorial note. The label prop:provability-quantifiers names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Quantifier axiom schemes ax:q1 / ax:q2. Same unit, quantifier axiom group ax:q1 / ax:q2. Each item is one scheme instance followed by the deduction theorem.

Source reference. Original reference ax-q1: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 520–520. Prerequisite pointer invoked at proof line(s) 1135; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-q2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 521–521. Prerequisite pointer invoked at proof line(s) 1134; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *1.* $A(t) \vdash \exists x\, A(x)$.

2. $\forall x\, A(x) \vdash A(t)$.

Proof. 1. By ax-q2 and the deduction theorem.

2. By ax-q1 and the deduction theorem.

□